

Active Learning (AL) - based Solutions

Presented by:
Prof. Nandan Sudarsanam
Aakash Sarraf
Nikshay Jain, Sujal Burad

Behavioural & Operational Features

General information:

- Claims history: contents + building.
- Previous insurance: PDS survey indicates 65–70% retention.
- Short-term letting: ~2% (Airbnb penetration UK).
- Business use: 5% of UK households.
- Unoccupied: 30/60/90-day patterns.
- Used `uniform()`, `triangular()`, `random.choices()` extensively
- Applied $\pm 15\%$ variance to tier premiums for natural distribution
- Introduced noise in correlations to avoid overfitting

Localization:

- 12% Insurance Premium Tax (UK-specific)
- £50K rounding for `sum_insured` (industry standard)
- Crime rates weighted by urban/rural (ONS data)
- Property age reflects UK's old housing stock (45% pre-1945)

Safety & Security:

- Smart home growth driven by insurtech partnerships
- Fire safety: UK Building Regulations mandate in new builds
- Security: London properties 68% adoption vs rural 42%

Premium Discounts:

- Smart home: -3% to -5%
- Fire safety: -5% to -8%
- Security systems: -2% to -4%

Summary Statistics Table

Feature	Mean	Median	Std Dev	Min	Max
premium_with_tax	£374	£358	£142	£156	£892
sum_insured	£368K	£350K	£128K	£140K	£900K
property_age	68 yrs	64 yrs	42 yrs	0	200
no_of_claims_5yr	0.24	0	0.52	0	3
premium_payment_behaviour	84.2	85	9.3	60	100
fraud_probability	0.089	0.052	0.098	0.001	0.847

Expected Correlations Validated:

1. ✓ Detached homes → Higher `sum_insured` ($r=0.78$)
2. ✓ Old properties → Higher premiums ($r=0.34$)
3. ✓ Claims history → Lower payment behaviour ($r=-0.42$)
4. ✓ Urban location → Higher crime rate ($r=0.61$)
5. ✓ Security systems → Lower premiums ($r=-0.18$)

Data Generation

- **Products:** Buildings, Contents, Combined, Landlord, etc.
- **Occupations:** Weighted by UK ONS data (retired 28%, professional 22%, etc.).
- **Property Types:** Terraced 26%, Semi 24%, Flats 24%, Detached 19%.
- **Urban/Rural:** 83% urban.
- **Risk Bands:** Flood/storm calibrated to EA risk maps.
- **Property Ages:** UK is old → 50% >75 years.
- Premium built from:
 - a. Tier-based base premium.
 - b. Property-type multipliers (Detached +25%, Terraced -5%).
 - c. Natural hazards applied via weighted surcharges.
- Sum insured based on **rebuild cost bands** and ONS housing cost indices.

```
df.columns.values
```

```
✓ 0.0s
```

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      'property_market_value', 'short_term_letting', 'fraud_probability',  
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```

Fraud Probability Model & Strategic enablers

- A fully non-linear, interpretable model based on UK fraud studies.
- Fraud likelihood grows **exponentially** with:
 1. High claim count in last 5 years
 2. Multiple payment defaults
 3. if property is labelled unoccupied + business_use is labelled as 'Y'
 4. high sum insured relative to market value ratio
 5. very low risk premium vs actual premium ratio
 6. Short-term letting + multiple claims.
- Penalizes **mismatch** between **contents vs building claims**.
- **Payment Behaviour:** Historical payment patterns predict future defaults and claim settlement risks.
- **Property Risk:** Physical characteristics and location exposure determine claim frequency and severity.
- **Coverage Anomaly:** Unusual coverage patterns, underwriting errors or policy misalignment with actual risk.
- **Demographic & Insurance History:** Customer profile, claim history for baseline risk & to predict future claims.

Impact on Premium

- 1. Properties >100 years old average 22% higher premiums
- 2. 10% of English homes in flood zones → 10% assigned risk 4-5
- 3. Weather claims exceeded £100M for 7 consecutive quarters
- 4. 170,000 storm claims in 2022 (£473M payout)
- 5. Urban: Weighted toward moderate risk (crime_rate 2-3)
Rural: Weighted toward low risk (crime_rate 1-2)
London averaged £333 premium (36% surge) vs rural £280
- 6. Occupation correlates with property value and risk profile -
Professionals pay 41% more than retired individuals.

Primary Research Sources

Source	Organization	Key Data Points
UK Home Insurance Market Report 2024	Mordor Intelligence	Property type distribution, premium ranges, regional variations
UK Home Insurance Statistics 2024	Mintel Store	Market size (£9.4B by 2029), ownership rates (65%), policy types
Household Insurance Market Dynamics	GlobalData	GWP growth (8.7% in 2023), premium increases, claims landscape
Home Insurance Statistics	Confused.com	Average premiums (£396 Q2 2024), claim frequencies, fraud rates
UK Property Claims Data	Uswitch	Weather claims (£585M in 2024), regional variations

Fraud correlation

- 1. Late additional coverage requests (>£15K) before claims are red flags
- 2. Airbnb properties with high claims flagged as fraud opportunity in industry reports
- 3. Commercial activities in residential properties elevate fire/liability risks
- 4. Properties unoccupied >30 days show 3.2× higher fraud rate
- 5. Triangular with mode at 85 (typical good payer)
 - I. Score <70: 15% of population, high fraud risk
 - II. Score 70-85: 50% of population, average risk
 - III. Score >85: 35% of population, low risk

Formulation

$$\text{fraud_score} = 0.30 \times \text{claims_risk} + 0.25 \times \text{payment_risk} + 0.20 \times \text{property_risk} + 0.15 \times \text{coverage_anomaly} + 0.05 \times \text{demographic_risk} + 0.05 \times \text{insurance_history}$$

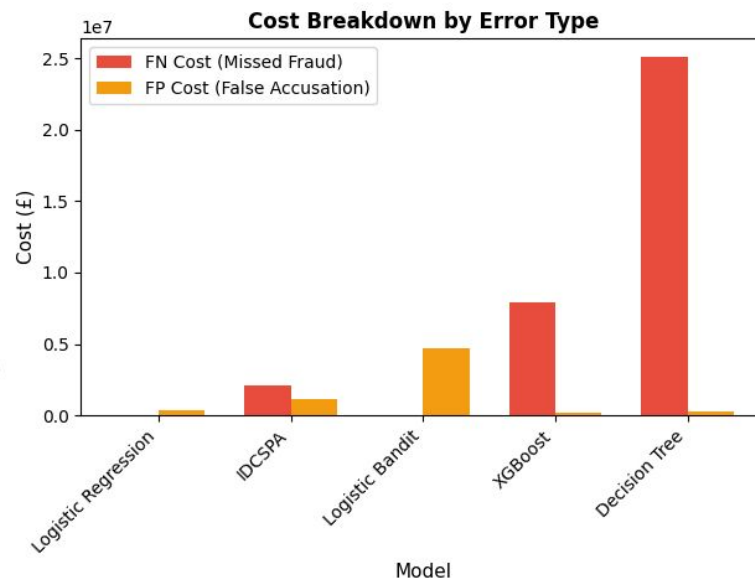
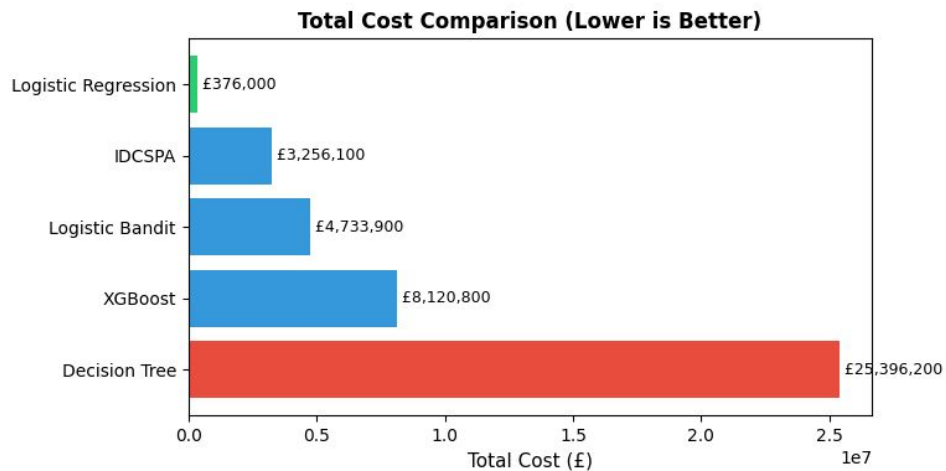
Where:

$$\begin{aligned} \text{claims_risk} &= \exp(2.5 \times \text{normalized_claims}) - 1 \\ \text{payment_risk} &= \exp(2.0 \times \text{normalized_poor_payment}) - 1 \\ \text{property_risk} &= \sin(\pi \times \text{natural_risk_combined} / 2) \times 1.5 \\ \text{coverage_anomaly} &= \exp(2 \times (\text{coverage_ratio} - 0.8)) - 1 \text{ [if ratio > 0.8]} \end{aligned}$$

$$\text{fraud_probability} = \text{sigmoid}((\text{fraud_score} - 2.5) \times 1.2)$$

Baseline ML model

Cost-Sensitive Fraud Detection Model Comparison



Active Learning Framework

1. **Rule-based filtering for suspicious top-K** : Initial screening using business rules to identify the highest-risk customers for manual review.
2. **Investigate flagged → true labels revealed** : Analysts verify flagged cases and provide ground-truth labels for training data.
3. **Train XGBoost with cost-sensitive loss.** : Model learns that missing bad cases (FN) costs more than false alarms (FP), so it prioritizes catching risky customers.
4. **Monthly incoming customers.** : New customer batch arrives each month requiring risk assessment.
5. **Query uncertain, high-risk, and small subset of low-risk:** Active learning selects cases where model is uncertain, obvious high-risk cases, and a few low-risk samples for labeling.
6. **Retrain using expanded labeled pool.** : New labels are added and the model is retrained to improve predictions on the growing dataset.
7. **Monthly evaluation on fixed test set.** : Consistent performance tracking using the same held-out test set each month.
8. **Track cost, recall, precision, FN cost, FP cost, and data efficiency:** Monitor financial impact alongside recall (prioritized due to high FN cost) and precision—total cost reduction is the goal.

Active Learning Query Strategy

Balanced Three-Stream Labeling:

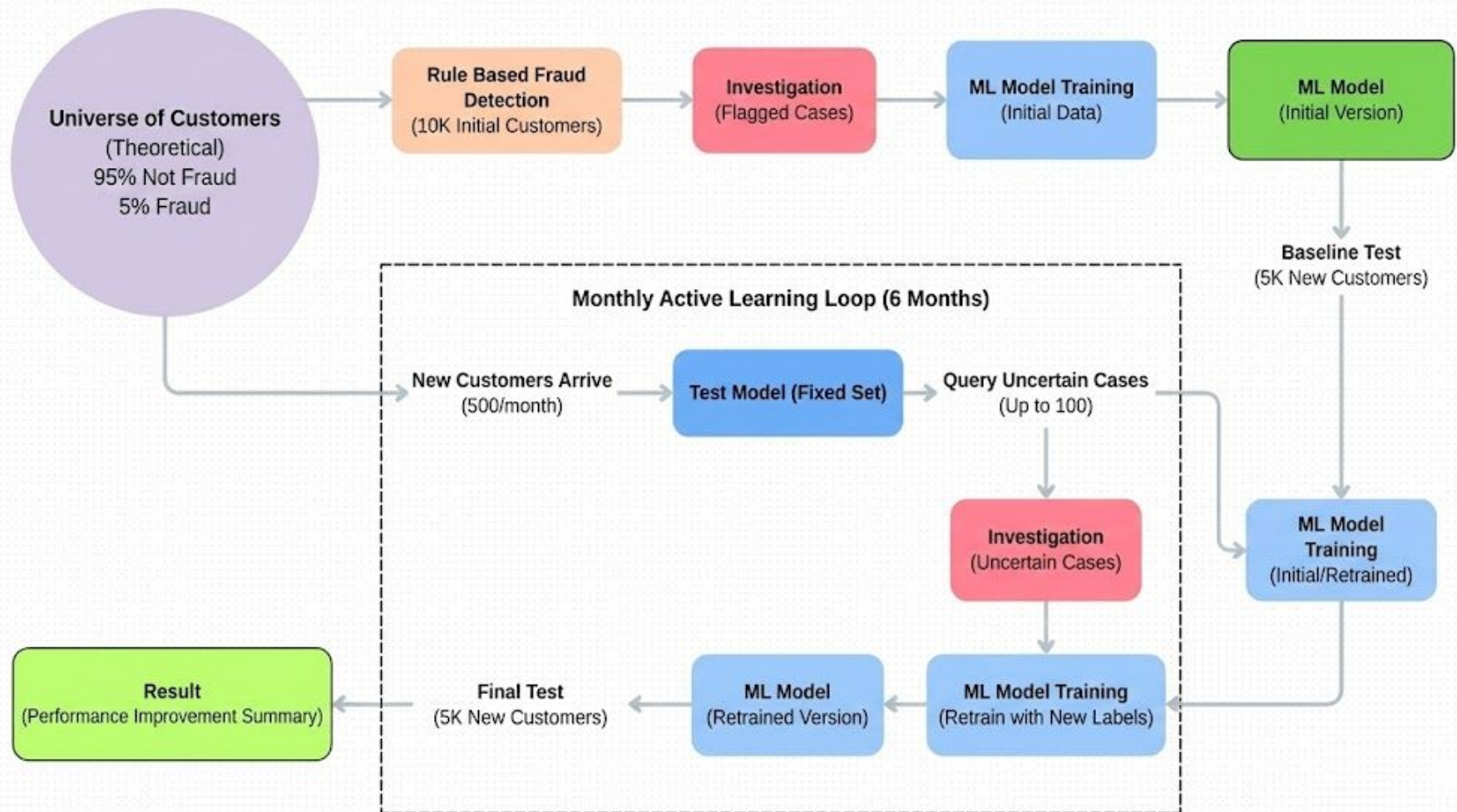
- **Uncertain Predictions (70% samples)**
 - Model is indecisive
- **High-Confidence Fraud (20% samples)**
 - Strong fraud signals that need verification → prevent costly false negatives
- **Low-Confidence Legitimate (10% samples)**
 - Edge cases that confuse the model → reduce algorithmic bias and false positives

Month-wise process:

1. New customers → samples of 500/1000/2000.
2. Predict probabilities → identify uncertain & high-risk.
3. Investigate top-N, in accordance with organisation's query budget.
4. Retrain model with expanded labels.
5. Evaluate on fixed test set to ensure consistency.

Strategy:

Cost-weighted sampling ensures efficient use of annotation budget

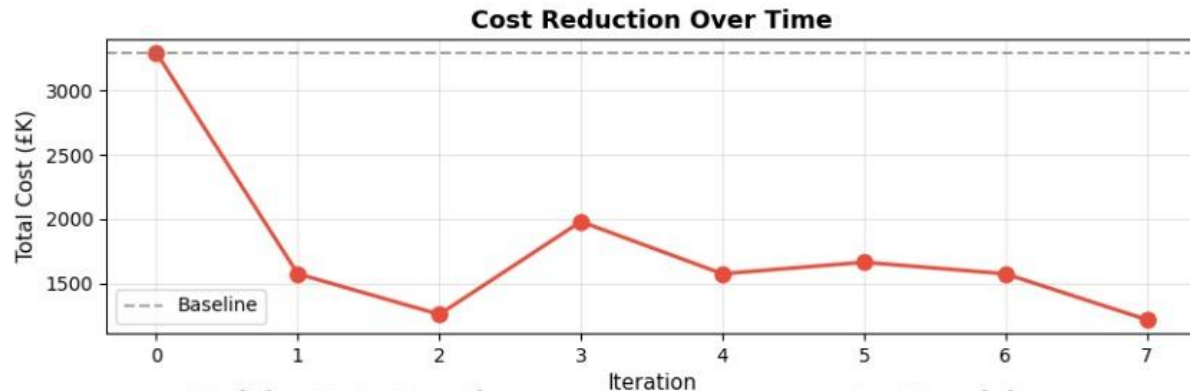


Performance Tracking

Tracked each period:

- **Total cost (£)** - FN cost + FP cost tracked per period. [Business impact: Direct cost measurement of AL decisions]
- **False Negative count (most expensive metric)** - Primary cost driver. [Justifies recall-focused optimization over precision]
- **Recall as primary metric** - minimizes costly false negatives. [Since FN cost >> FP cost, recall prioritized for model validation].
- **Precision & threshold evolution** - cost-sensitive calibration. [Tracks how model threshold adjusts as labeled dataset grows]
- **Labeled dataset growth & cost per sample** - AL efficiency metric. [Measures whether active learning reduces total cost vs baseline]
- **Cost Reduction vs baseline** - final performance indicator. [Quantifies value delivered by cost-sensitive AL strategy]

AL Performance



Metric	Baseline	After 6M	Change
Total Cost (£)	3,291,700	1,220,200	+62.9%
Recall (%)	96.4	97.9	+1.5pp
Precision (%)	80.0	81.8	+1.8pp
FN Count	10	5	+5

PERFORMANCE IMPROVEMENT SUMMARY

Baseline (Initial Model):

Cost: £3,291,700

Recall: 96.4%

Final (After 6 months AL):

Cost: £1,220,100

Recall: 97.9%

Improvement:

Cost Reduction: +62.9%

Recall Gain: +1.5 percentage points

Total Labeled Data: 2093 samples

UK Data Sources Used

Population, Occupation, Housing:

- ONS Labour Force Survey
- UK Housing Survey 2023
- Gov.uk English Housing Data

Risk Maps:

- Environment Agency Flood Risk Map
- Met Office Severe Weather Patterns

Insurance Market:

- Association of British Insurers (ABI) Home Insurance Report
- Insurance Fraud Bureau (IFB) Fraud Typology
- FCA Household Insurance Market Analytics

Payment Behaviour:

- Experian UK Financial Wellness Index
- FCA Consumer Risk Survey

Academic & Industry Sources

1. **Mordor Intelligence** (2024). *UK Home Insurance Market Report*.
<https://www.mordorintelligence.com/industry-reports/uk-home-insurance-market>
 - Property type distribution, premium ranges
2. **Mintel** (2024). *UK Home Insurance Market Report 2024*.
<https://store.mintel.com/report/uk-home-insurance-market-report-2024>
 - Market size (£9.4B forecast), ownership rates
3. **GlobalData** (2024). *UK Household Insurance Market Dynamics*.
 - GWP growth, premium inflation, claims landscape
4. **Confused.com** (2024). *UK Home Insurance Statistics*.
<https://www.confused.com/home-insurance/home-insurance-statistics>
 - Average premiums (£396), claim frequencies
5. **Uswitch** (2025). *Home Insurance Statistics*.
<https://www.uswitch.com/home-insurance/studies/home-insurance-statistics-uk/>
 - Weather claims (£585M), regional variations
6. **Association of British Insurers (ABI)** (2024). *Insurance Fraud Statistics*.
 - 10-15% fraud rate benchmark
7. **Office for National Statistics (ONS)** (2021). *UK Census - Housing Stock*.
 - Property age distribution, urban/rural split
8. **English Housing Survey** (2023). *Housing Stock Report*.
 - Property condition, age bands

Regulatory & Government Sources

9. **Financial Conduct Authority (FCA)** (2024). *Home Insurance Pricing Practices*.
 - Auto-renewal regulations, price walking reforms
10. **Environment Agency** (2024). *Flood Risk Assessment*.
 - 10% of English homes in flood zones